2026

PROJECTED PROJECTS

A Journal

JANUARY

Projected Projects

A Journal

January 2026

2026 by J. Wyatt Biggs with Scribus
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Preface

PRESSED FLOWERS

Taylor Bennett

Project Facts

Time

Labor	25-30 M
Harvesting	10 M
Setup	10 M
Pressing	13 W

Tools

Paper Towels

Style: No Pattern

Cardboard

Style: Corrugated
Dim: 1'H x 3'W x 1/4"T
Origin: Moving box, flattened

Yearbook

Style: Hardcover
Dim: 11"H x 8.5"W
Qty: 3

Storage Tote

Weight: 20-25 lbs
Dim: 2'L x 1'W x 1'H
Material: Plastic, filled w/ Magazines

Materials

Flowers

Location: Colorado Springs, CO
Season: Summer?

1 **Name, Common:** Oxeye Daisy
Name, Scientific: *Leucanthemum vulgare*
Qty: 6

2 **Name, Common:** Black-eyed Susan
Name, Scientific: *Rudbeckia fulgida*
Qty: 5

...

Materials, Continued

Flowers, Continued

3 **Name, Common:** English Lavender
Name, Scientific: *Lavandula angustifolia*
Qty: 4

4 **Name, Common:** Narrow-Mouth Penstemon
Name, Scientific: *Penstemon breviculus*
Qty: 4

5 **Name, Common:** Garden Rose
Name, Scientific: *(Rosa x hybridia)*
Qty: 2

Introduction

BIRTHDAY CARDS

Katie Wilson

Project Facts

Time

Labor 1-2 H

Tools

Pen

Mfr: ?
Model: ?
Type: Felt tip
Color: Black
D: ?

Colored Pencils

Mfr: PrismaColor
Color: *Assorted*

Water Colors

1 **Mfr:** Koi
Color: Copper
Style: Sparkly

2 **Mfr:** Arteza
Color: *Assorted*

Paint Brushes

Mfr: Artist's Loft
Type: Rounded tip
Size: Medium-small, like the size of a blunted Sharpie

Scissors

Mfr: ?
Model: ?

Materials

Water Color Paper

Mfr: Artist's Loft
Color: White
Dim: 9"H x 6"W
Weight: similar to cereal box cardboard

...

Materials, Continued

Stickers

Mfr: Peter Pauper Press
Model: The Simple Life Cottagecore
Style: *Assorted artistic*
Type: *Assorted content- Seasons, flowers, animals*
Form: Shiny, smooth texture; not worried about tears
Color: Brownish background; transparent

Introduction

3D PRINTED DRAGONS

J. Wyatt Biggs

Project Facts

Time

Labor	1 H
Machine	13 H
Idle	12 H
Active	1 H

Tools

3D Printer

Mfr: Bambu Lab
Model: X1C

3D Printer Auxiliary

Mfr: Bambu Lab
Model: AMS

Materials

Filament

Plastic: PLA
Mfr: MatterHackers
Form: Spool
D: 1.75mm
Type: Silky MH Build Series Magic

1 **PN:** M-2U9-SZN8
Color 1: Gold
Color 2: Magenta
Qty: ~100 g

2 **PN:** M-AAK-X697
Color 1: Blue
Color 2: Green
Qty: ~100 g

3 **PN:** M-247-UEQE
Color 1: Pink
Color 2: Blue
Qty: ~100 g

Introduction

Because a handful of friends wanted an articulated dragon as a toy, dragons of different sizes and colors were printed in dual-color filament. The quality of the articulation was of interest due to the potential use of articulating components in novel and currently unimagined products. The physical properties and appearance of dual-color filaments was of interest as it results in a some-what iridescent surface finish which could prove useful in projects with an ornamental rather than purely functional aim.

Method

Setup

Instead of designing an articulated dragon myself, Taylor Bennett picked out the model [1]. I chose to use exotic filaments since

Method-Setup Continued

these prints were ornamental rather than functional. I limited the filament choices to those found on the MatterHackers website [2].

First Dragon

To get a feel for the surface finish and appropriate slicing settings, I printed a test Benchy [3 and Fig. 3.1] with BambuLab's default PLA settings. I then began printing the dragon at 75% scale with the Gold-Magenta PLA filament but it halted about half-way through due to a fan speed issue. This was resolved by replacing the complete hotend assembly [4].

Figure 3.1: Gold-Magenta Benchy

The first fully printed dragon had a defect: while printing the first few layers, one of the dragon's links detached from the build plate and imbedded itself into another portion of the dragon, fusing several links together (hereafter referred to as an inclusion defect). Printing a second time with the same settings produced the first dragon without defects [Fig. 3.2].

Figure 3.2: Final print of the first dragon

Second Dragon

The second dragon was printed at 125% scale with the Blue-Green PLA filament. The first attempt had an inclusion defect. The second attempt was successfully printed

Figure 3.3 (Left): First attempt at printing the second dragon, with inclusion defect; Figure 3.4 (right): Final print of the second dragon on a shelf

without defects using an initial layer height of 0.3mm and adaptive cubic infill.

Third and Fourth Dragon

Preserving the print settings from the second dragon, the third dragon was printed at 75% scale with the Pink-Blue PLA filament. The first attempt suffered from an inclusion defect which in resulted the dragon's head falling off as well as fused links. The infill was changed from adaptive cubic to gyroid for the second attempt. The second attempt produced another inclusion defect that fused several links together. No settings were changed before the third attempt and the print suffered the same defect as the second attempt. The build plate was removed, handwashed over a sink with water and handsoap, dried with a flour sack towel, and placed back in the X1C. The two proceeding prints had no defects.

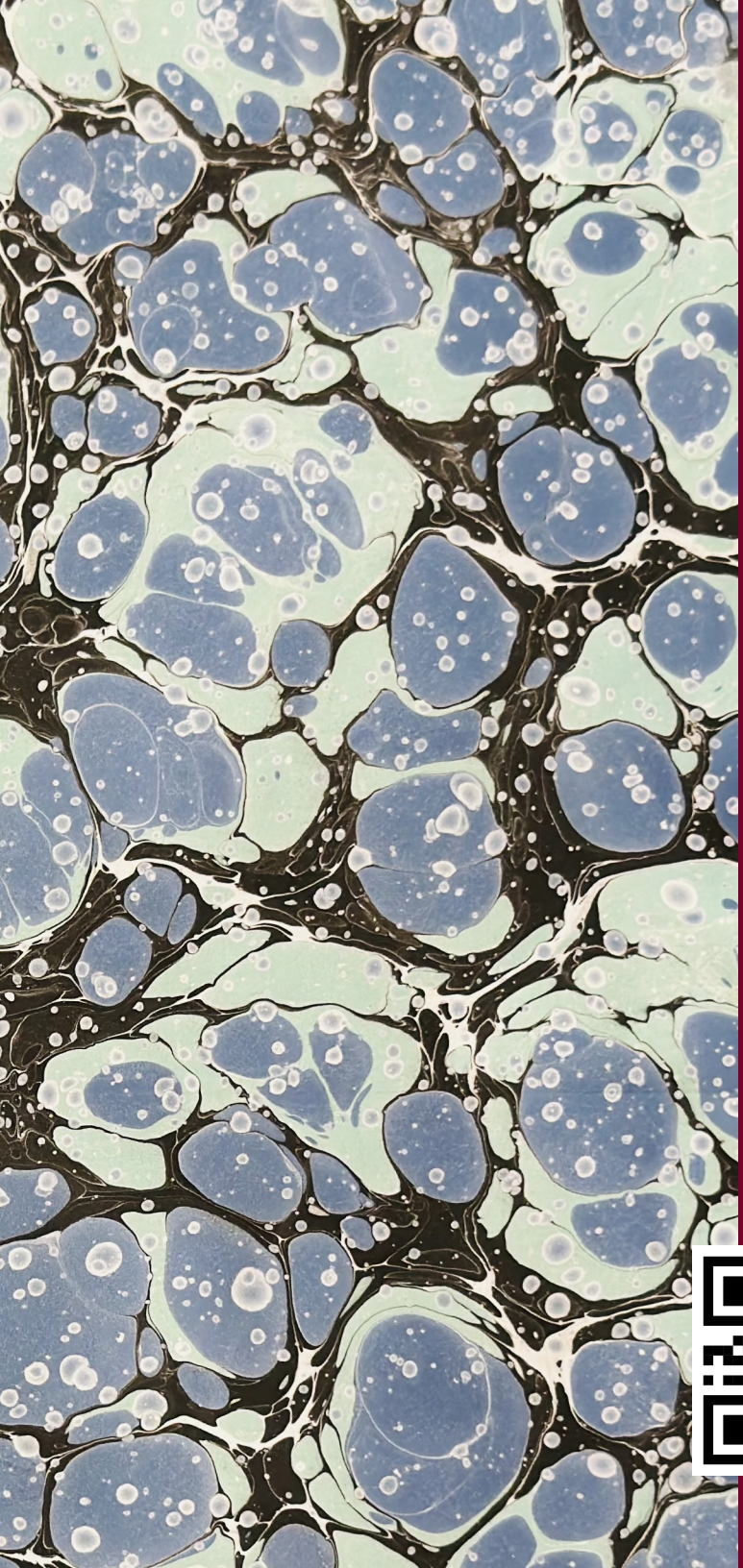
Figure 3.5: Final prints of the third (side) and fourth (side) dragon

Results

The articulation of each final dragons is fully functional and behaves as predicted, having the most freedom about the pitch axis, followed by yaw and then roll. No tools were used to measure the physical properties, but the dual-colored PLA has physical properties roughly equivalent to that of single-colored PLA. The Pink-Blue PLA filament produces in the most visually interesting surface finish, followed by a tie between the Blue-Green PLA filament and the Gold-Magenta PLA filament. The dragons printed at 75% scale provide a more enjoyable fidgeting experience compared to the dragon printed at 125% scale.

Discussion

Because the model was sourced from MakerWorld rather than designed from scratch, this project demonstrates only that small-scale, chain-like articulations are feasible; meaningful modifications beyond simple scaling are not possible without developing a fully parametric model. The mechanical differences between single- and dual-colored filament would require a load frame and standard testing methods for 3D prints which would likely be similar to existing tensile strength tests (ASTM D638/ISO 527) [5] but would have to account for the anisotropy of the 3D printed test coupons.

References

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[2] https://www.matterhackers.com/store/c/3d-printer-filament
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